

A. Arduino Board Base Connections

- Connect the **5V pin** to your power positive (+) rail. (This is for the LED, Buzzer, and RFID only. **Do NOT power the servo from here**).
- Connect a **GND pin** to your common ground rail.

B. MFRC522 RFID Reader

The RFID module uses the SPI interface. Follow these connections precisely, as they are non-negotiable for proper SPI communication.

1. **VCC (or 3.3V):** Connect **ONLY to Arduino 3.3V**. *Warning: Connecting this pin to 5V will permanently damage the RFID module.*
2. **RST (Reset):** Connect to Arduino Pin **D9** (#define RST_PIN 9).
3. **GND (Ground):** Connect to common ground.
4. **IRQ:** Not used in this code.
5. **MISO:** Connect to Arduino Pin **D12** (Standard SPI pin).
6. **MOSI:** Connect to Arduino Pin **D11** (Standard SPI pin).
7. **SCK (Serial Clock):** Connect to Arduino Pin **D13** (Standard SPI pin).
8. **SDA (or SS/CS):** Connect to Arduino Pin **D10** (#define SS_PIN 10).

C. Continuous Rotation Servo

You must use an external battery pack for this component. A 4 x AA battery pack is ideal (providing ~4.8V - 6V). The standard color codes for servos are usually Red, Black, and White/Orange.

1. **Signal/Data (White or Orange wire):** Connect to Arduino Pin **D6** (#define SERVO_PIN 6).
2. **Power (+) (Red wire):** Connect to the **Positive (+) terminal of the external battery pack**. *Do not connect this to the Arduino 5V pin.*
3. **Ground (-) (Black or Brown wire):** Connect to the **common ground rail**. (The negative terminal of the external battery pack must also be connected here).

D. Buzzer

For the **tone()** function in the code to work correctly, this should be a passive piezo buzzer.

1. **Positive (+) terminal:** Connect to Arduino Pin **D7** (#define BUZZER_PIN 7).
2. **Negative (-) terminal:** Connect to common ground.

E. Access LED

We must include a current-limiting resistor to protect the LED.

1. **Anode (Positive, long leg):** Connect to Arduino Pin **D8** (#define LED_PIN 8).
2. **Cathode (Negative, short leg):** Connect to one side of a **220Ω resistor**.
3. Connect the other side of the **220Ω resistor** to common ground.